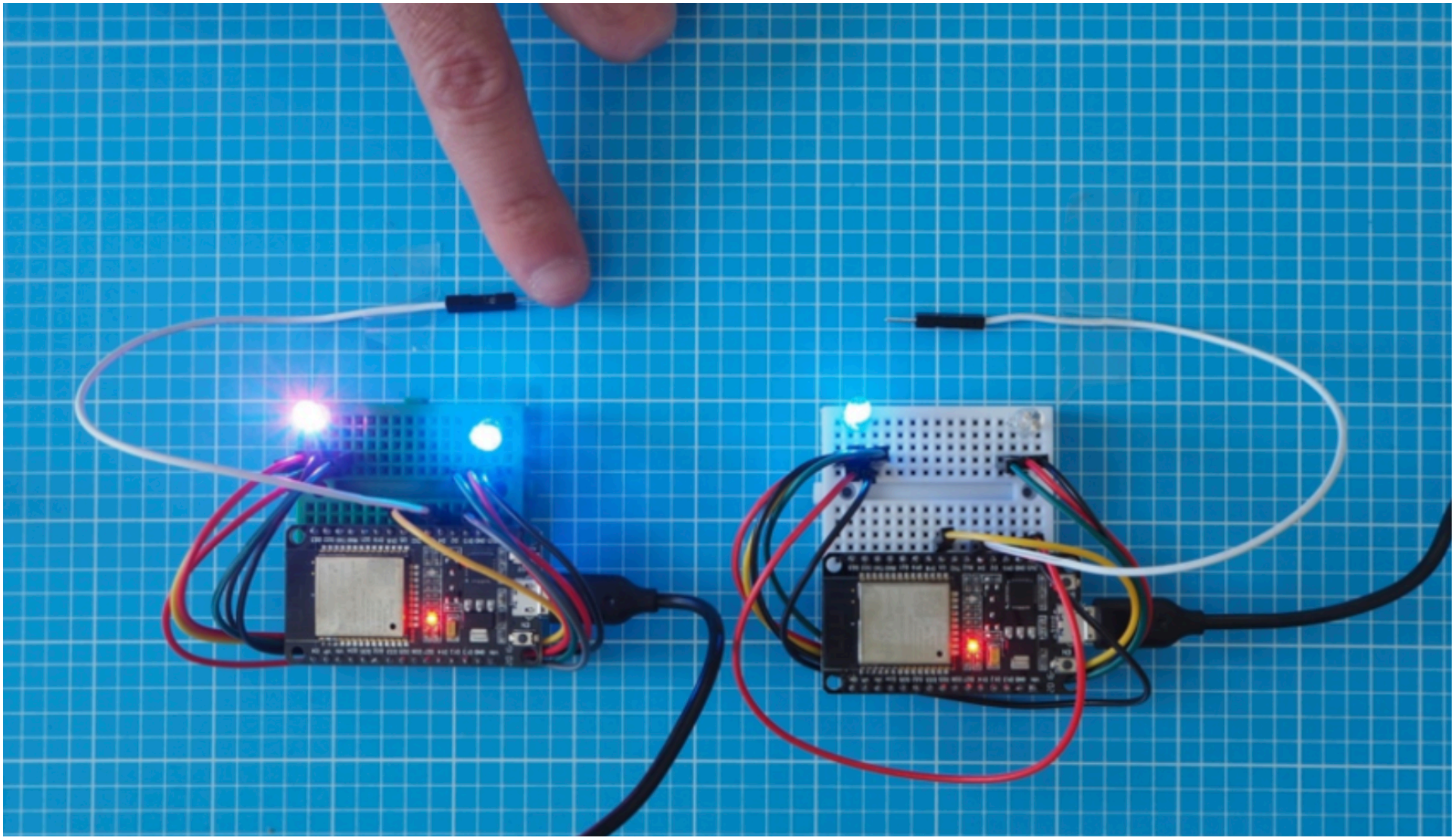
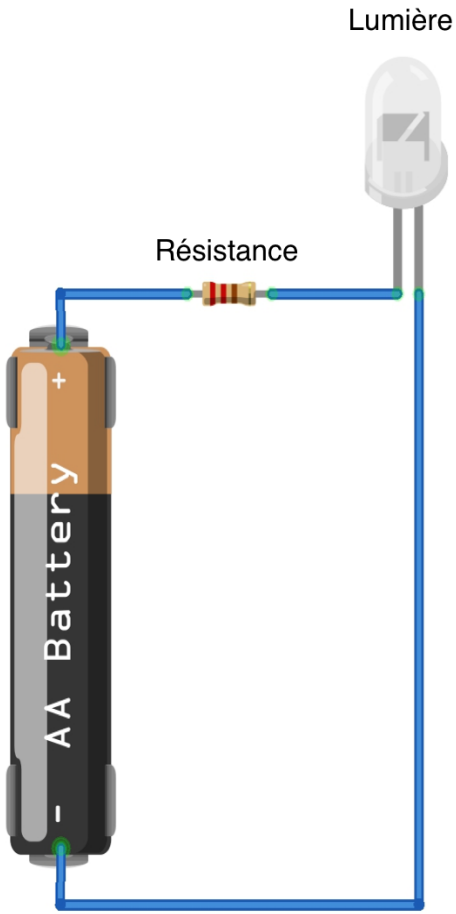
EDM2641 - Médias tangibles

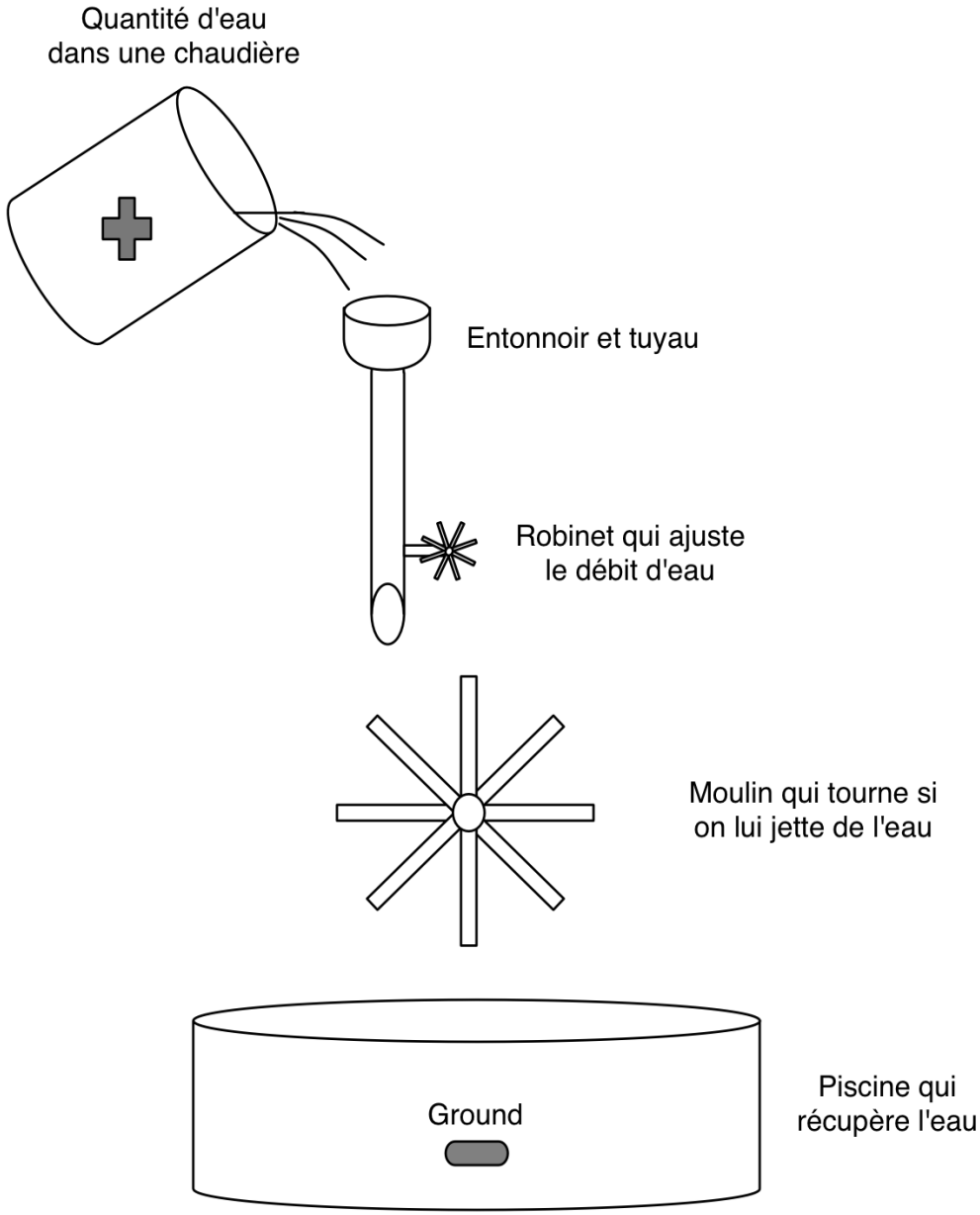
Professeur: Patrice Coulombe



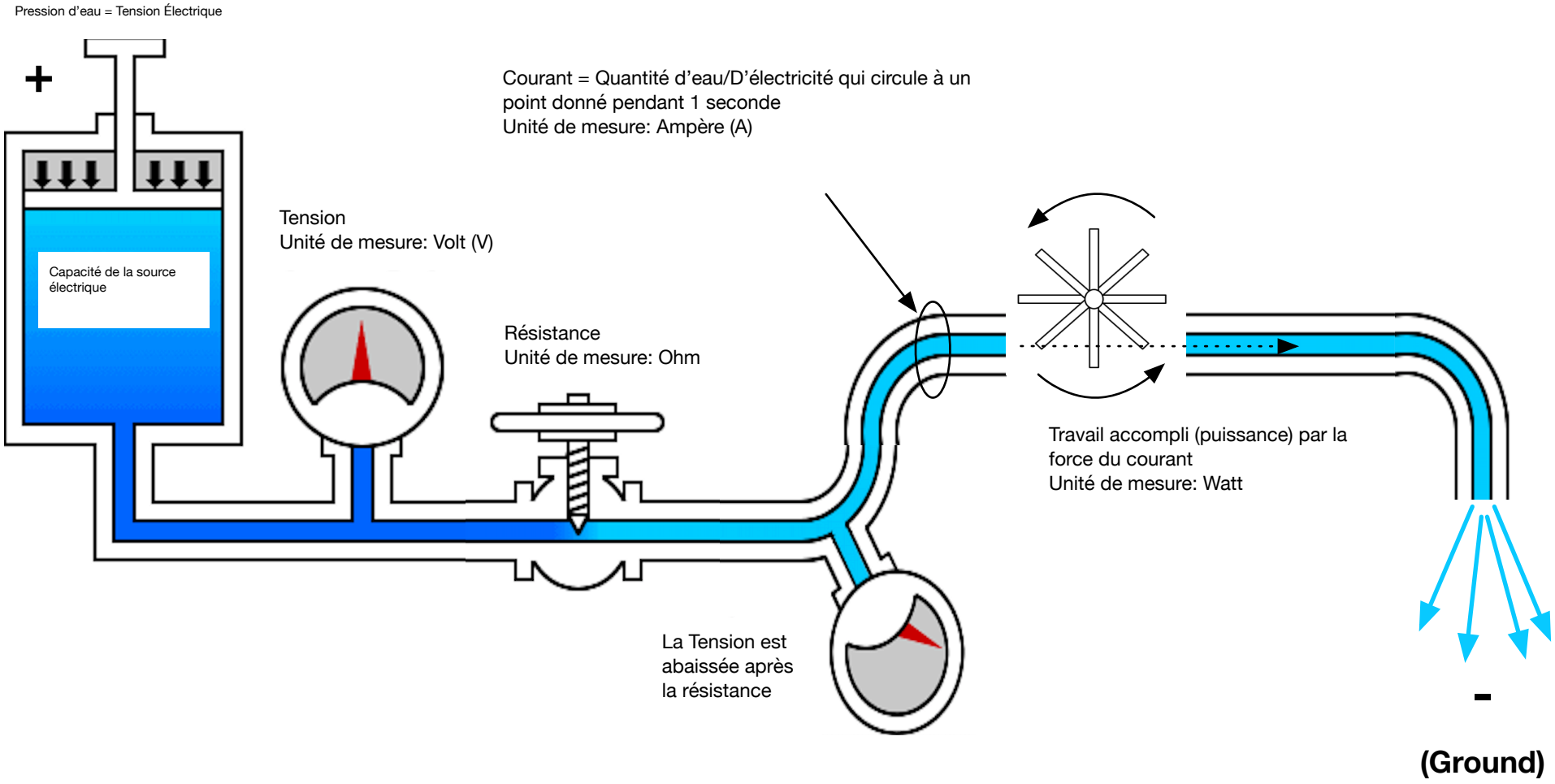
Semaine 2: Introduction à l’électronique et Arduino

Analogie: Moulin à eau vs Réseau électrique

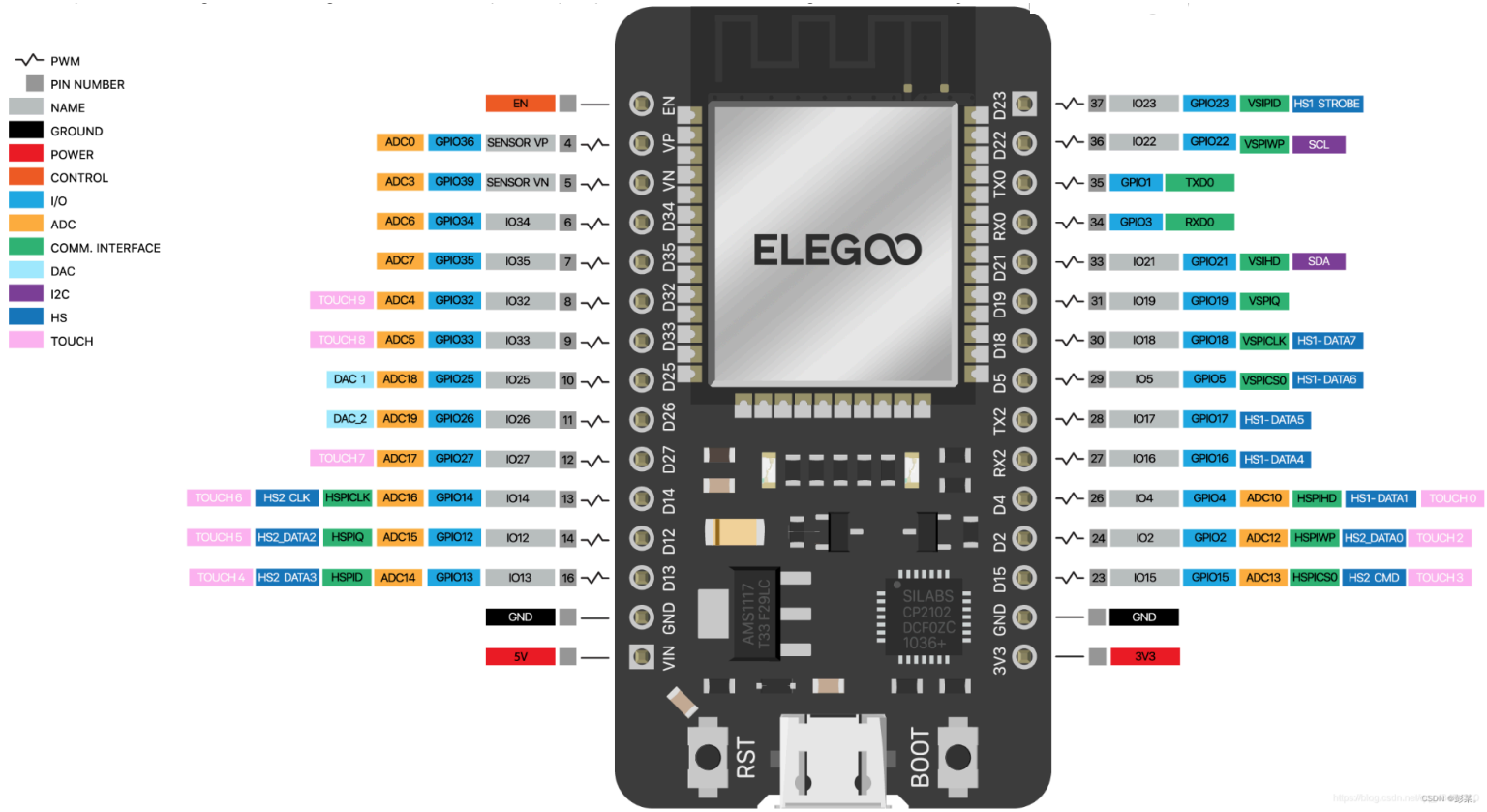


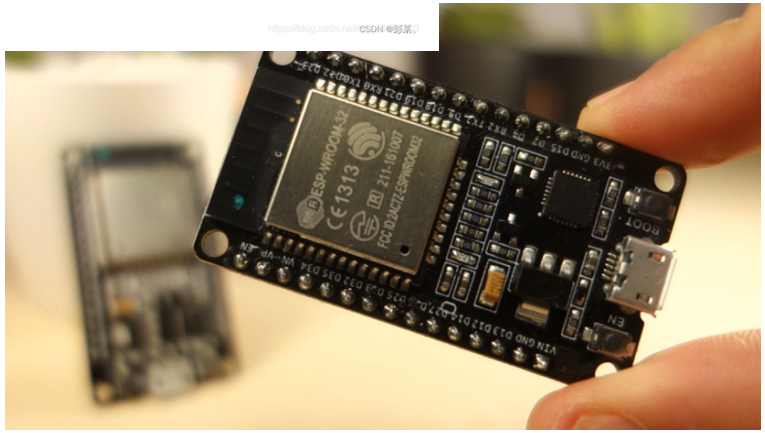


Analogie: Réseau aqueduc vs réseau électrique

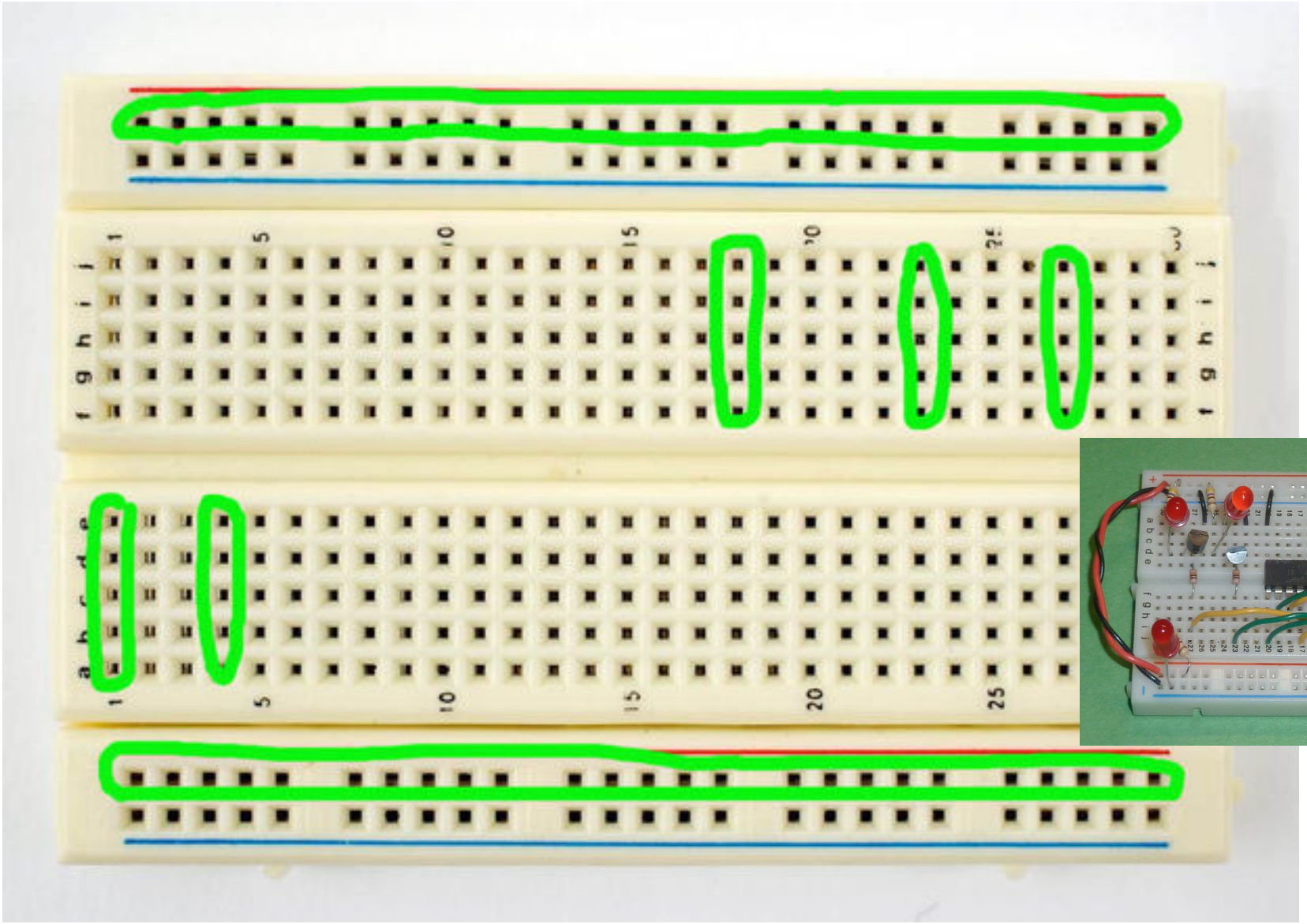


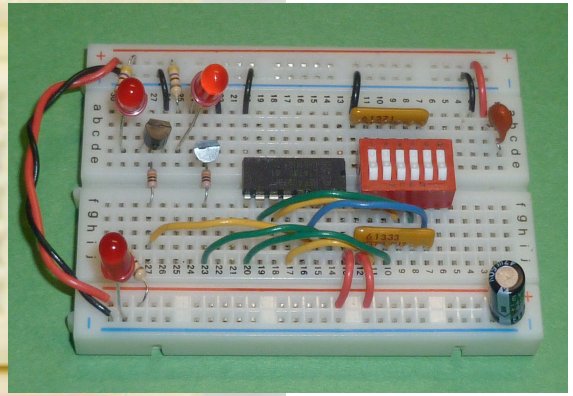
ESP32 (DOIT devkit v1 30pins) : Microntrôleur programmable



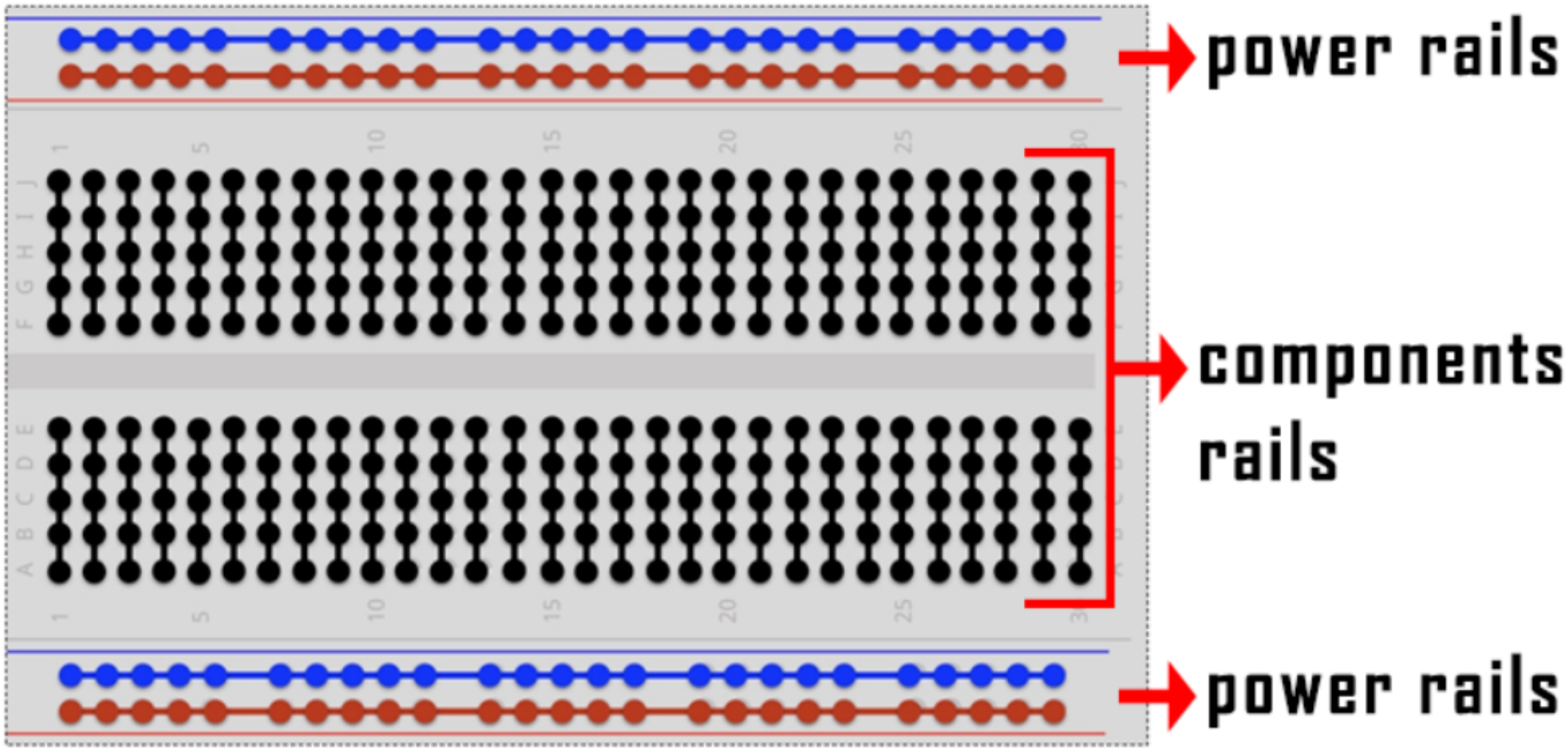


Plaque de prototypage (“Breadboard”)

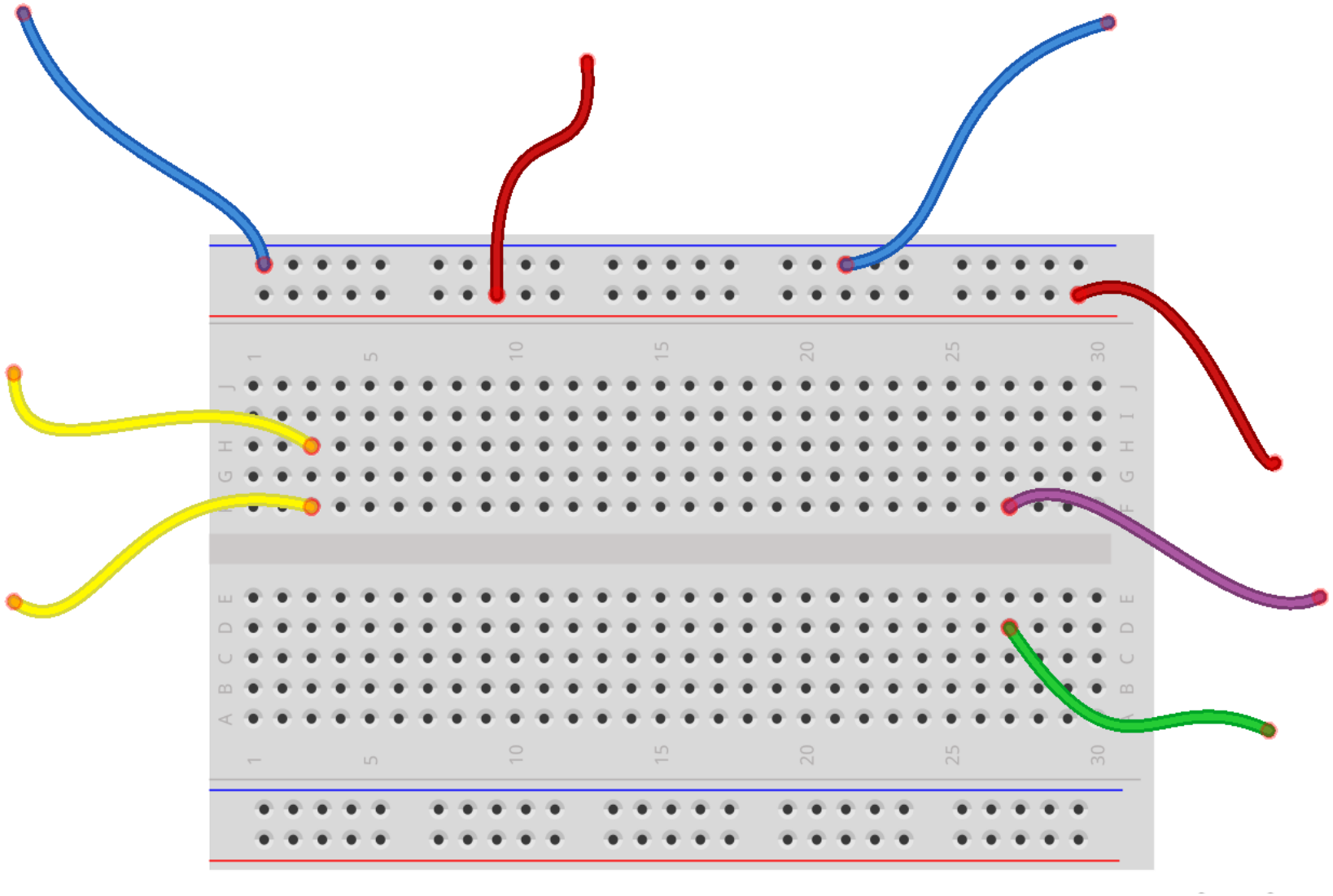


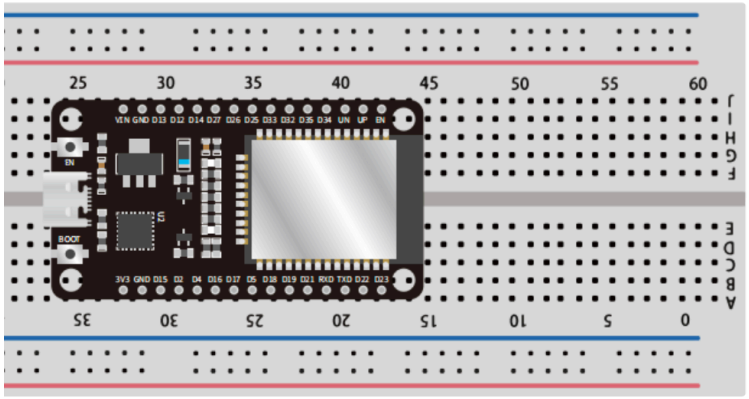


Plaque de prototypage (“Breadboard”)

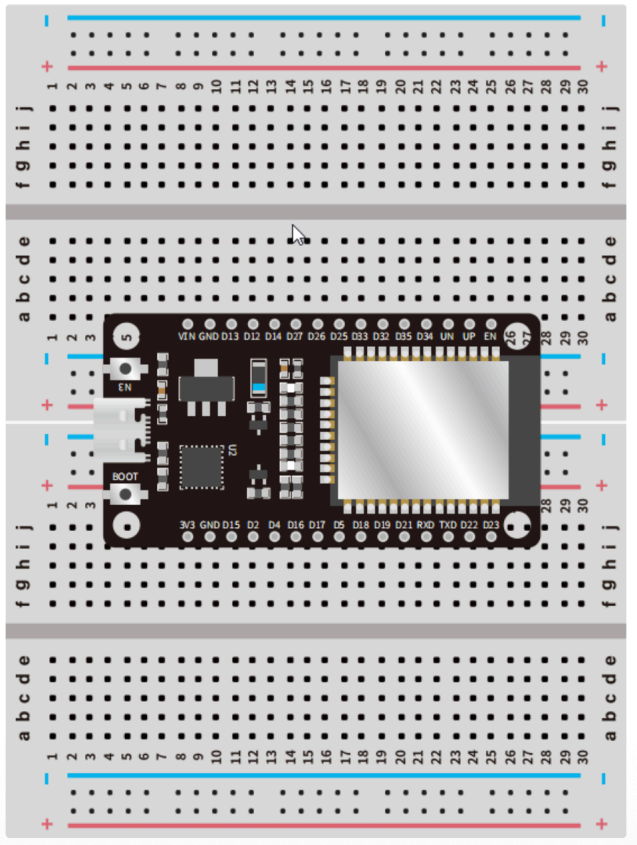


Sauriez-vous dire quels fils sont connectés avec quels fils?

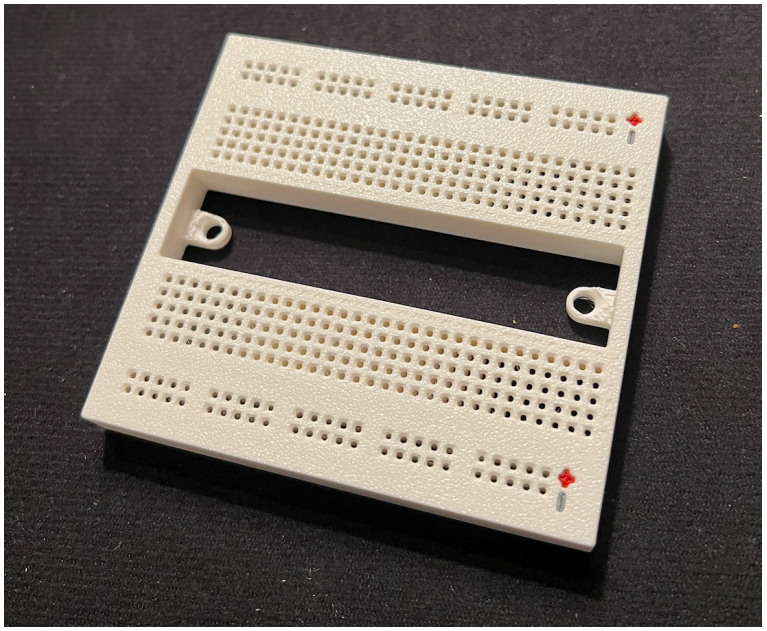




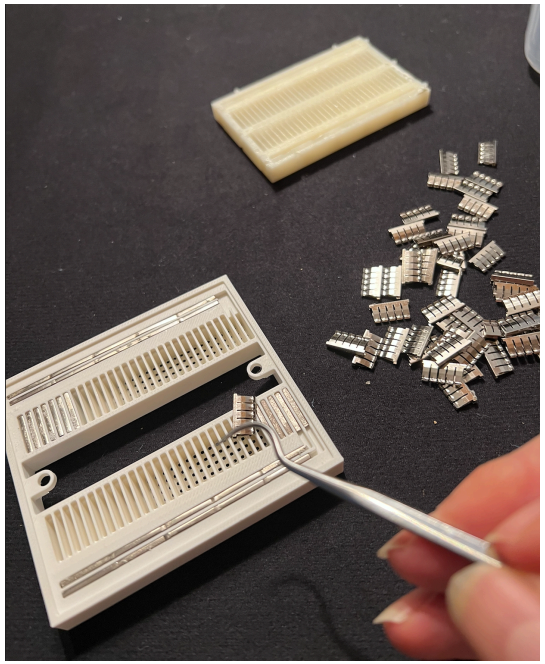
Le ESP32 prend trop de place sur un seul breadboat.
Les pins qui sont connectées à celui-ci sont cachés par lui



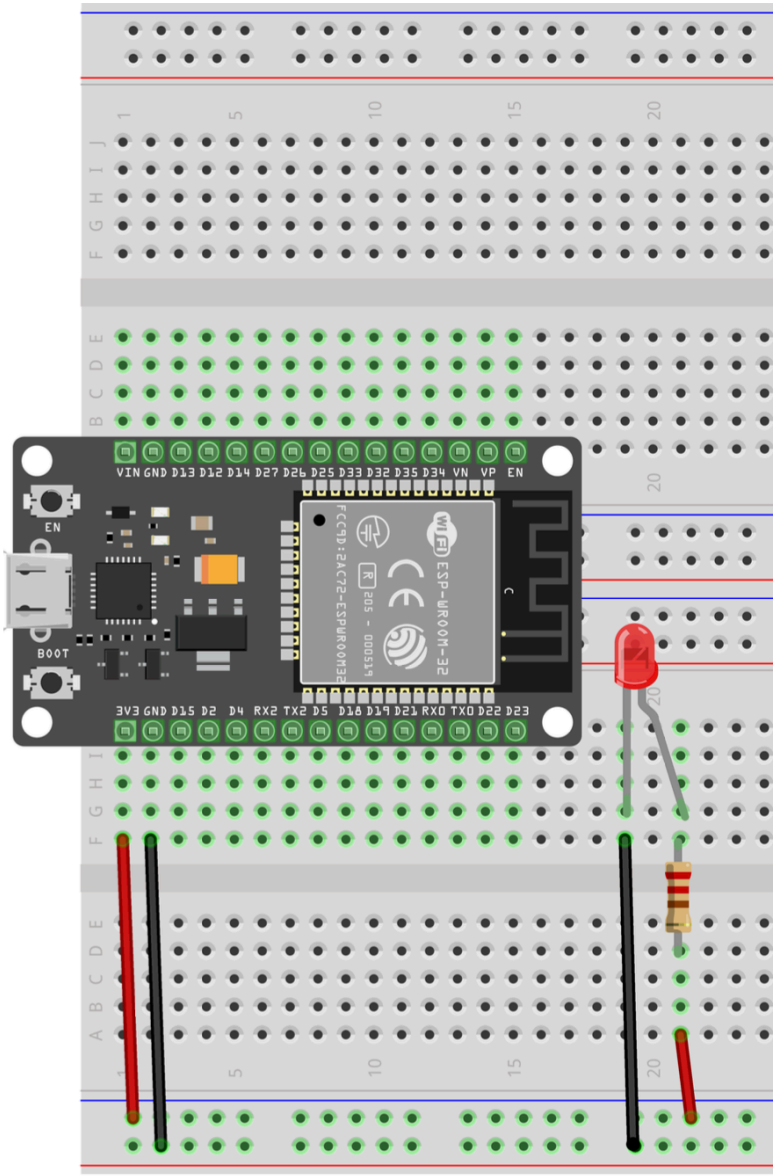
Il est necessaire d’assembler les deux breadboards de cette façon afin d’avoir accès aux connections.



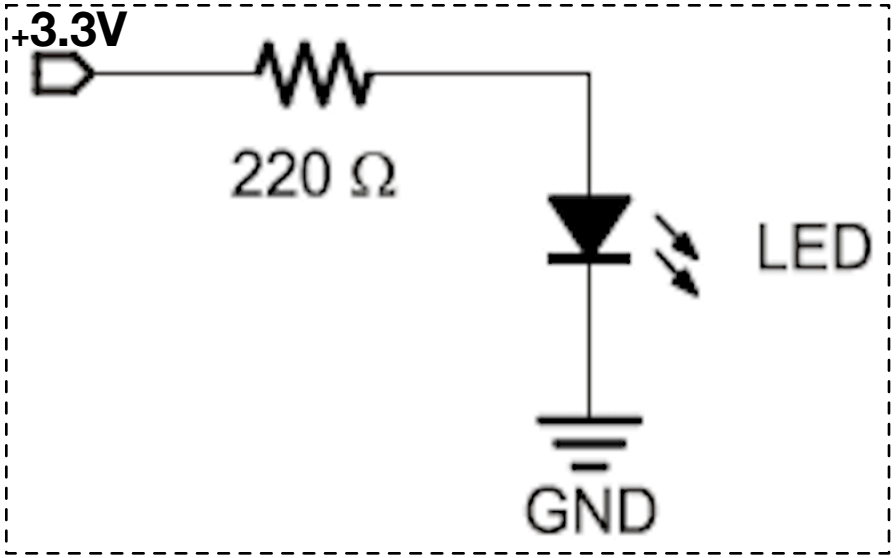
Vous pourriez imprimer ce breadboard qui est conçu spécialement pour intégrer un ESP32… (makerworld.com)

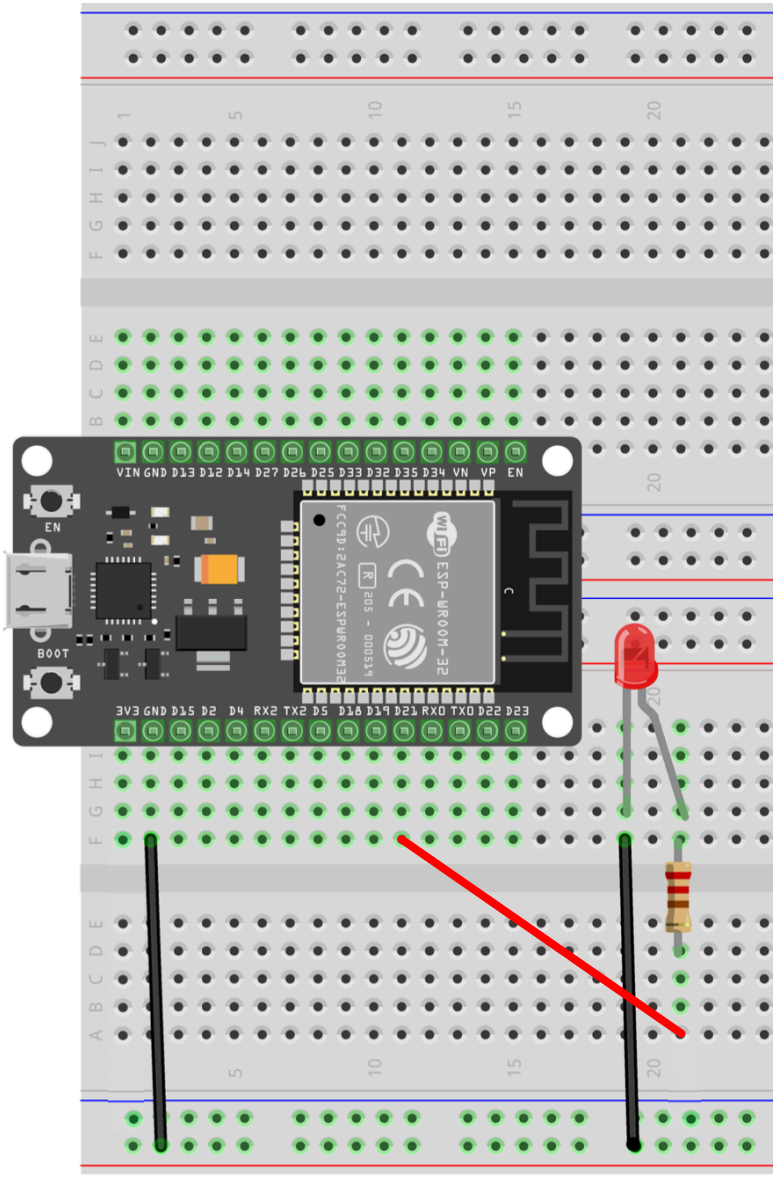


Mais, vous devrez transférer tous les connecteurs d’un de vos breadboards provenant du kit vers celui-ci!

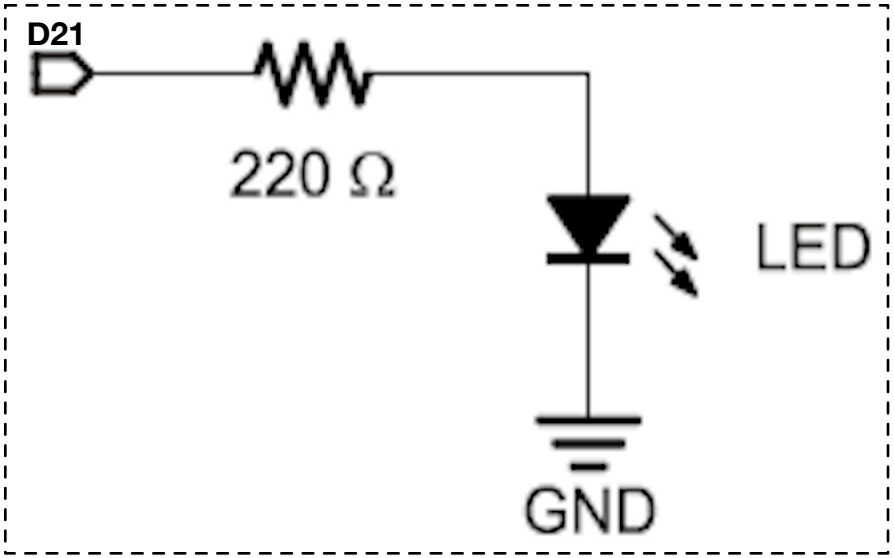


Circuit LED branché dans le 3.3V





Circuit LED branché dans D21



Sortie nom arguments (quand il y en a...)

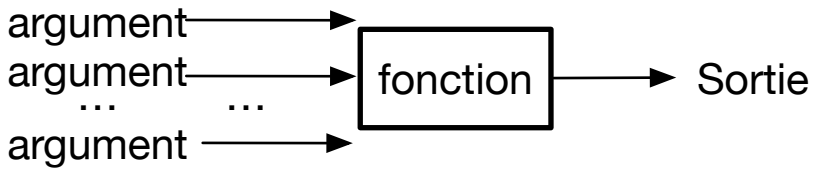
```
void setup() {
  // put your setup code here, to run once:

}

void loop() {
  // put your main code here, to run repeatedly:

  int x = sum(10, 20);
}
```

Utilisation de la fonction sum()

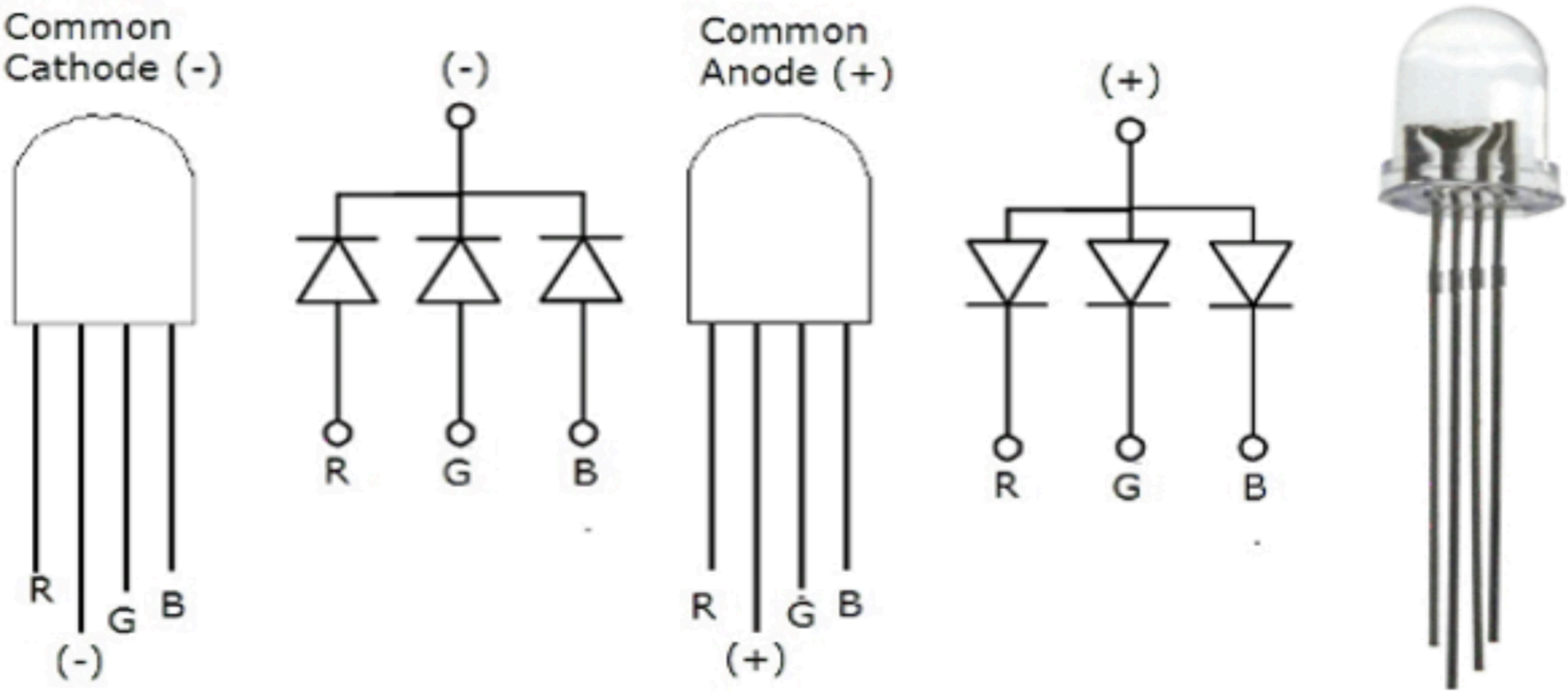
Sortie nom arguments (Il y en a deux!)

```
int sum(int a, int b)
{

  return a + b;

}
```

Ce qui est envoyé à la sortie



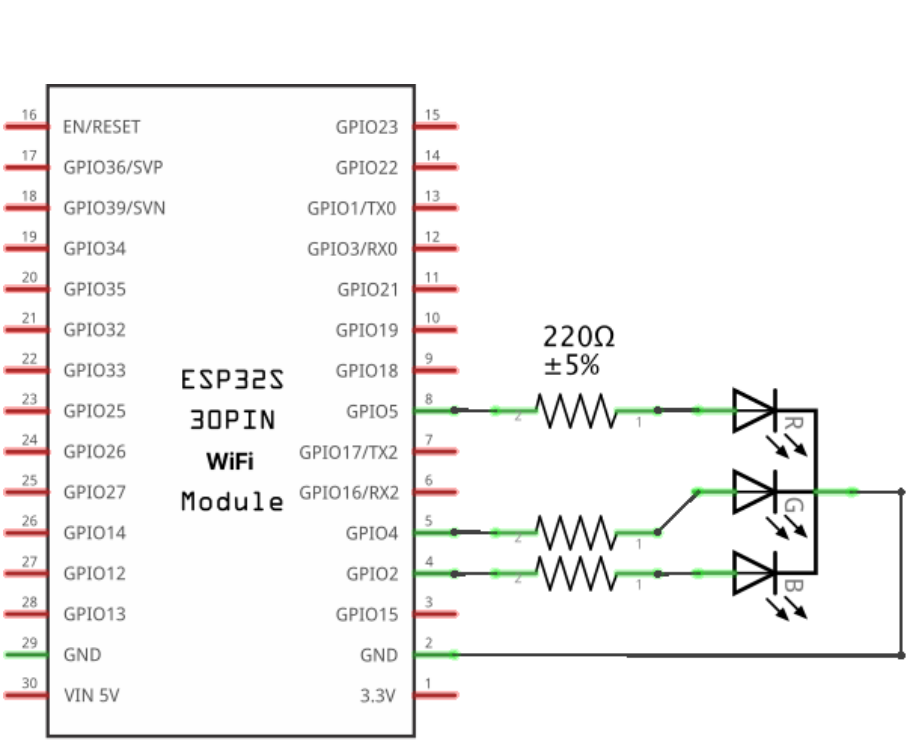
```
1  int pinLed = 21;                  //pin IO de la LED
2
3  void setup()
4  {
5    pinMode(pinLed, OUTPUT);     //pin en mode "sortie"
6  }
7
8  void loop()
9  {
10   digitalWrite(pinLed, HIGH); //allume la pin
11   delay(1000);                 //pause de 1 seconde
12
13   digitalWrite(pinLed, LOW); //éteint la pin
14   delay(1000);                //pause de 1 seconde
15 }
```

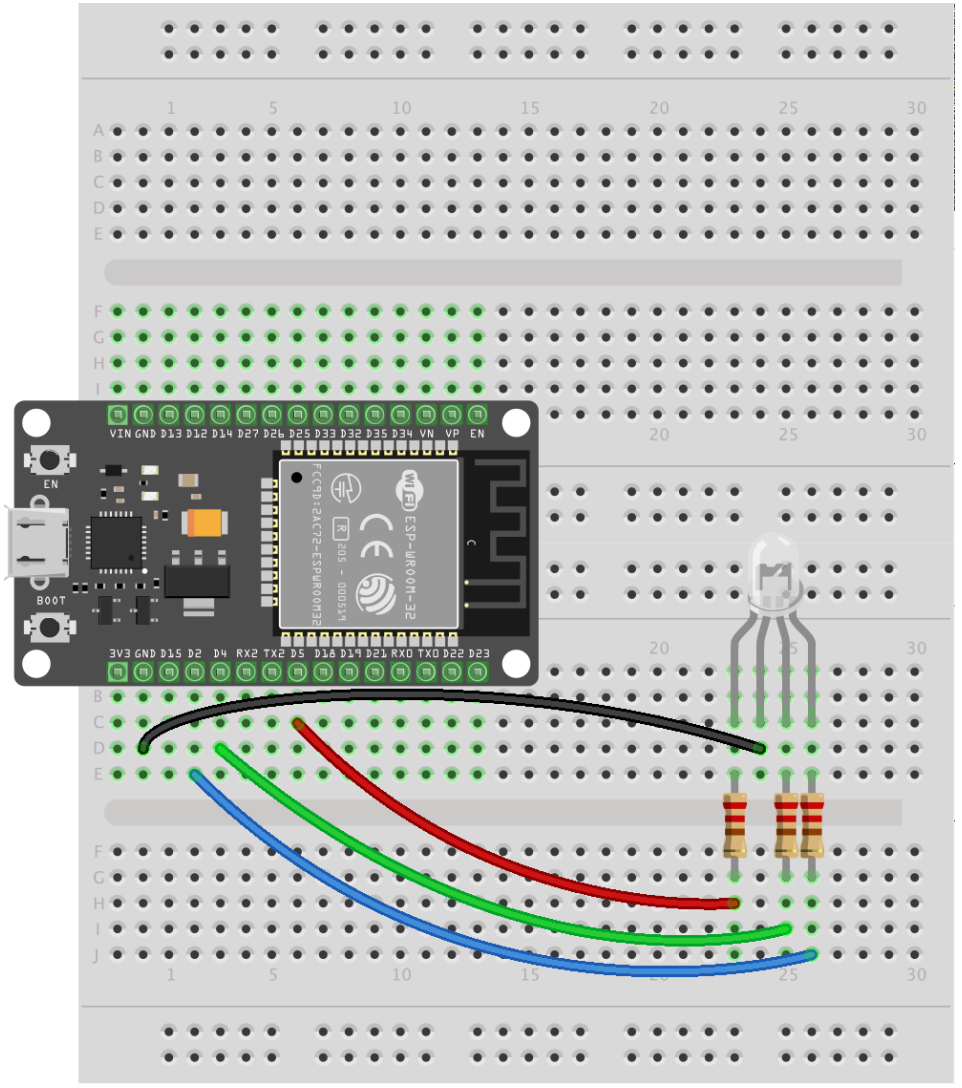


Dans votre kit, c'est un DEL "Common Cathode" que vous avez

… mais prévoyez qu'un jour vous rencontrerez des LED "Common Anode…

Circuit LED rgb





Programme pour une LED RGB

Sortie on/off = digitalWrite()

Alternance entre les couleurs

```
1  int pinRouge =  2;
2  int pinVert  =  4;
3  int pinBleu  =  5;
4
5  void setup()
6  {
7    pinMode(pinRouge, OUTPUT);
8    pinMode(pinVert,  OUTPUT);
9    pinMode(pinBleu,  OUTPUT);
10 }
11
12 void loop()
13 {
14   digitalWrite(pinRouge, LOW);
15   digitalWrite(pinVert,  LOW);
16   digitalWrite(pinBleu,  LOW);
17   delay(500);
18
19   digitalWrite(pinRouge, HIGH);
20   digitalWrite(pinVert,  LOW);
21   digitalWrite(pinBleu,  LOW);
22   delay(500);
23
24   digitalWrite(pinRouge, LOW);
25   digitalWrite(pinVert,  HIGH);
26   digitalWrite(pinBleu,  LOW);
27   delay(500);
28
29   digitalWrite(pinRouge, LOW);
30   digitalWrite(pinVert,  LOW);
31   digitalWrite(pinBleu,  HIGH);
32   delay(500);
33 }
```

Sortie graduée = analogWrite()

```
1  const int pinRouge =  2;
2  const int pinVert  =  4;
3  const int pinBleu  =  5;
4
5  void setup()
6  {
7    pinMode(pinRouge, OUTPUT);
8    pinMode(pinVert,  OUTPUT);
9    pinMode(pinBleu,  OUTPUT);
10 }
11
12 void loop()
13 {
14   for(int i = 0; i < 256; i++)
15   {
16     analogWrite(pinRouge, i);
17     delay(10);
18   }
19
20   // Exercise:
21   // Comment faire un fondu enchaîné entre les couleurs?
22   // ex: du rouge au vert, du vert au bleu et du bleu au rouge
23 }
```

```ino
int pinRouge =   2;
int pinVert  =   4;
int pinBleu  =   5;

void setup()
{
  pinMode(pinRouge, OUTPUT);
  pinMode(pinVert,  OUTPUT);
  pinMode(pinBleu,  OUTPUT);
}

void loop()
{
  for(int i = 0; i < 256; i++)
  {
    analogWrite(pinRouge, i);
    analogWrite(pinVert, 255 - i);
    delay(10);
  }
  for(int i = 0; i < 256; i++)
  {
    analogWrite(pinBleu, i);
    analogWrite(pinRouge, 255 - i);
    delay(10);
  }
  for(int i = 0; i < 256; i++)
  {
    analogWrite(pinVert, i);
    analogWrite(pinBleu, 255 - i);
    delay(10);
  }
}
```

Programme pour une LED RGB

Fondu enchaîné entre les couleurs //// Solution de l'exercise

SPOILER!

… Solution de l'exercise